Topic: Electrical series circuits

Question: If a battery supplies a constant voltage of 20 V, has an inductance of 2 h and a resistance of $10\ \Omega$, and assuming $i(0) = 0$, find the function that models the current.

Answer choices:

A $i(t) = 1 - e^{-5t}$

B $i(t) = 2 - 2e^{-5t}$

C $i(t) = 1 - e^{-10t}$

D $i(t) = 2 - 2e^{-10t}$

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Solution: B

This series circuit doesn't have a capacitor, which means we can remove that term from Kirchhoff's second law.

$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t)$$

$$L\frac{di}{dt} + Ri = E(t)$$

Plugging in what we know, we get

$$2\frac{di}{dt} + 10i = 20$$

$$i' + 5i = 10$$

This is a linear differential equation with integrating factor

$$I(t) = e^{\int P\ dt}$$

$$I(t) = e^{\int 5\ dt}$$

$$I(t) = e^{5t}$$

Therefore, we can rewrite the linear equation as

$$e^{5t}\frac{di}{dt} + 5e^{5t}i(t) = 10e^{5t}$$

$$\frac{d}{dt}(e^{5t}i(t)) = 10e^{5t}$$

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$$\int \frac{d}{dt}(e^{5t}i(t))\ dt = \int 10e^{5t}\ dt$$

$$e^{5t}i(t) = 2e^{5t} + C$$

$$i(t) = 2 + Ce^{-5t}$$

Substituting the initial condition $i(0) = 0$, we find

$$0 = 2 + Ce^{-5(0)}$$

$$0 = 2 + C(1)$$

$$C = -2$$

So we can write the equation modeling the current i over time t as

$$i(t) = 2 - 2e^{-5t}$$

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Topic: Electrical series circuits

Question: If a battery supplies a constant voltage of 18 V, has an inductance of 3 h and a resistance of $12\ \Omega$, and assuming $i(0) = 0$, find the current after 60 seconds.

Answer choices:

A 4.41

B 4.50

C 1.50

D 1.47

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Solution: C

This series circuit doesn't have a capacitor, which means we can remove that term from Kirchhoff's second law.

$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t)$$

$$L\frac{di}{dt} + Ri = E(t)$$

Plugging in what we know, we get

$$3\frac{di}{dt} + 12i = 18$$

$$i' + 4i = 6$$

This is a linear differential equation with integrating factor

$$I(t) = e^{\int P\ dt}$$

$$I(t) = e^{\int 4\ dt}$$

$$I(t) = e^{4t}$$

Therefore, we can rewrite the linear equation as

$$e^{4t}\frac{di}{dt} + 4e^{4t}i(t) = 6e^{4t}$$

$$\frac{d}{dt}(e^{4t}i(t)) = 6e^{4t}$$

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$$\int \frac{d}{dt}(e^{4t}i(t))\ dt = \int 6e^{4t}\ dt$$

$$e^{4t}i(t) = \frac{3}{2}e^{4t} + C$$

$$i(t) = \frac{3}{2} + Ce^{-4t}$$

Substituting the initial condition $i(0) = 0$, we find

$$0 = \frac{3}{2} + Ce^{-4(0)}$$

$$C = -\frac{3}{2}$$

So we can write the equation modeling the current i over time t as

$$i(t) = \frac{3}{2} - \frac{3}{2}e^{-4t}$$

$$i(t) = \frac{3}{2}(1 - e^{-4t})$$

Assuming time is measured in seconds, the current after 60 seconds is

$$i(1) = \frac{3}{2}(1 - e^{-4(60)})$$

$$i(1) = \frac{3}{2}(1 - e^{-240})$$

$$i(1) = 1.5$$

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Topic: Electrical series circuits

Question: In an electrical circuit, a battery supplies a constant voltage of 20 V; the circuit has an inductance of 4 h, a resistance of $3\ \Omega$, and capacitance of 2 f. Given $q(0) = 2$ and $i(0) = 0$, find the charge $q(t)$ on the capacitor.

Answer choices:

A $q(t) = 2e^{\frac{1}{2}t}(3 - e^{\frac{3}{4}t})$

B $q(t) = 2e^{-\frac{1}{4}t}(2 - e^{-\frac{1}{4}t})$

C $q(t) = -76e^{-\frac{1}{4}t} + 38e^{-\frac{1}{2}t} + 40$

D $q(t) = 76e^{-\frac{1}{4}t} - 38e^{-\frac{1}{2}t} + 40$

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Solution: C

This series circuit has a capacitor, which means we use Kirchhoff's second law.

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

Plugging in what we know, we get

$$4\frac{d^2q}{dt^2} + 3\frac{dq}{dt} + \frac{1}{2}q = 20$$

$$4q'' + 3q' + \frac{1}{2}q = 20$$

$$8q'' + 6q' + q = 40$$

We'll replace $g(x) = 40$ with $g(x) = 0$ to change the nonhomogeneous equation into a homogeneous equation

$$8q'' + 6q' + q = 0$$

then we'll solve the associated characteristic equation.

$$8r^2 + 6r + 1 = 0$$

$$(2r + 1)(4r + 1) = 0$$

$$r = -\frac{1}{2},\ -\frac{1}{4}$$

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Because we found distinct real roots, we know the circuit is overdamped. With distinct real roots, we can say that the complementary solution will be

$$q_c(x) = c_1 e^{-\frac{1}{4}t} + c_2 e^{-\frac{1}{2}t}$$

Since $g(x) = 40$ is a constant, we'll use A as our guess for the particular solution.

$$q_p(x) = A$$

Taking the first and second derivatives of the guess gives

$$q_p'(x) = 0$$

$$q_p''(x) = 0$$

Plug the derivatives into the original differential equation.

$$8(0) + 6(0) + A = 40$$

$$A = 40$$

Then the particular solution is

$$y_p(x) = 40$$

and the general solution is

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = c_1 e^{-\frac{1}{4}t} + c_2 e^{-\frac{1}{2}t} + 40$$

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The solution and its derivative are given by

$$q(t) = c_1 e^{-\frac{1}{4}t} + c_2 e^{-\frac{1}{2}t} + 40$$

$$q'(t) = i(t) = -\frac{1}{4}c_1 e^{-\frac{1}{4}t} - \frac{1}{2}c_2 e^{-\frac{1}{2}t}$$

Substituting the initial conditions $q(0) = 2$ and $i(0) = 0$ into these equations, we get

$$2 = c_1 e^{-\frac{1}{4}(0)} + c_2 e^{-\frac{1}{2}(0)} + 40$$

$$-38 = c_1 + c_2$$

and

$$0 = -\frac{1}{4}c_1 e^{-\frac{1}{4}(0)} - \frac{1}{2}c_2 e^{-\frac{1}{2}(0)}$$

$$0 = -\frac{1}{4}c_1 - \frac{1}{2}c_2$$

$$0 = c_1 + 2c_2$$

$$c_1 = -2c_2$$

Substituting this value into the first equation gives

$$-38 = -2c_2 + c_2$$

$$-38 = -c_2$$

$$c_2 = 38$$

Plugging this back into the equation for c_1, we get

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$$c_1 = -2(38)$$

$$c_1 = -76$$

So the general solution modeling charge q on the capacitor over time t is

$$q(t) = c_1 e^{-\frac{1}{4}t} + c_2 e^{-\frac{1}{2}t} + 40$$

$$q(t) = -76e^{-\frac{1}{4}t} + 38e^{-\frac{1}{2}t} + 40$$

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